



Building a Better Feedback Loop

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Dagster Labs,
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Time comes for us all

Entropy



Ozymandias

This is what happens to everything we build. Stone, software, data pipelines entropy comes for all of it. The only question is how fast, and whether you see it coming.

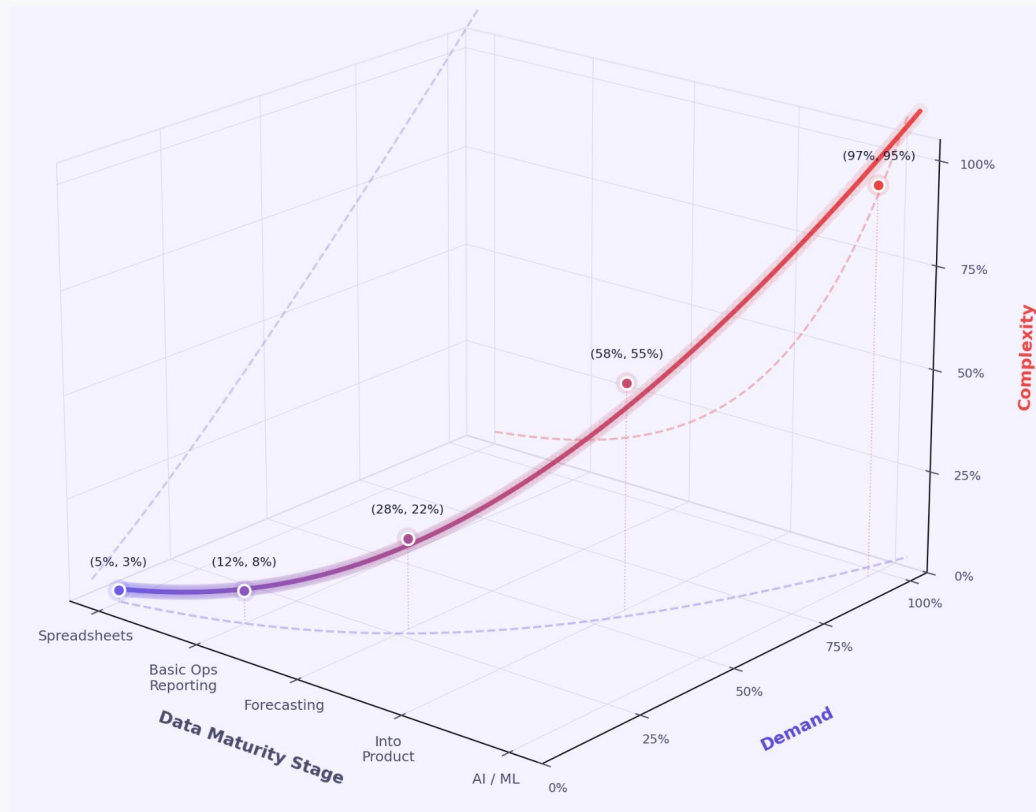
The Demand for Data



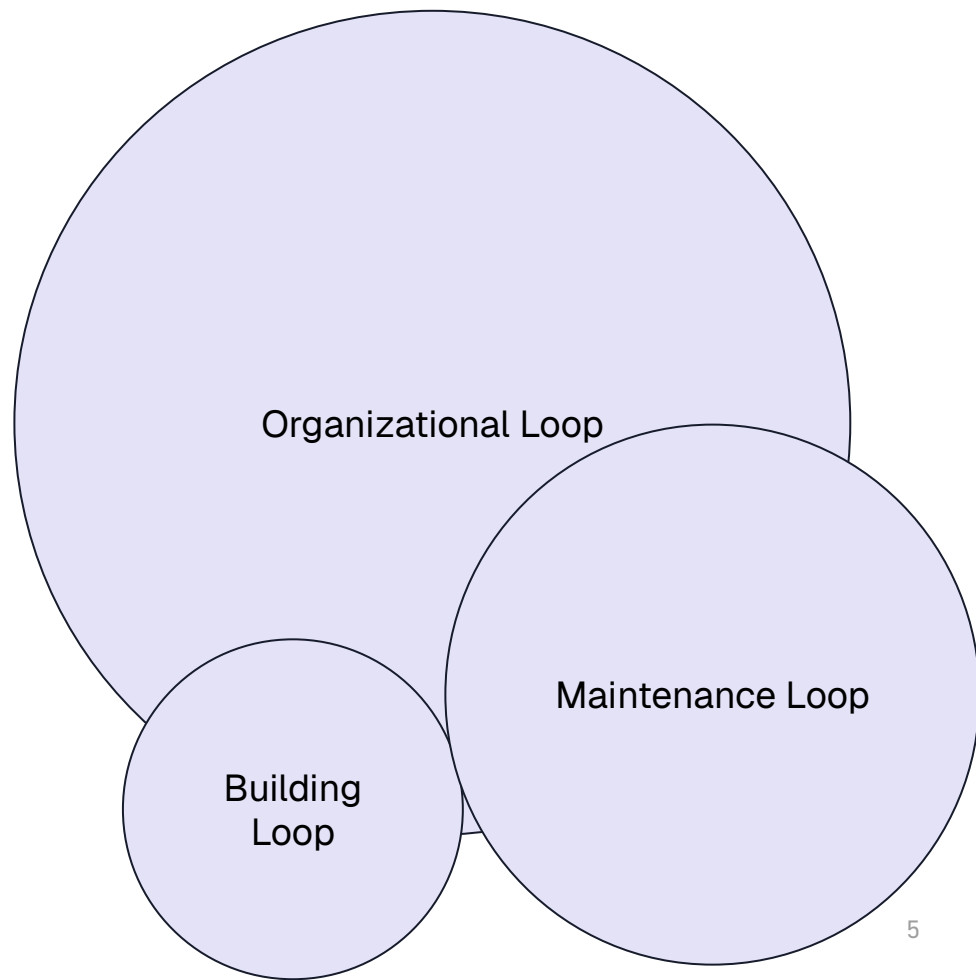
Organizational
Dynamics



Highly Contextual



Its **Feedback Loops** All the way down



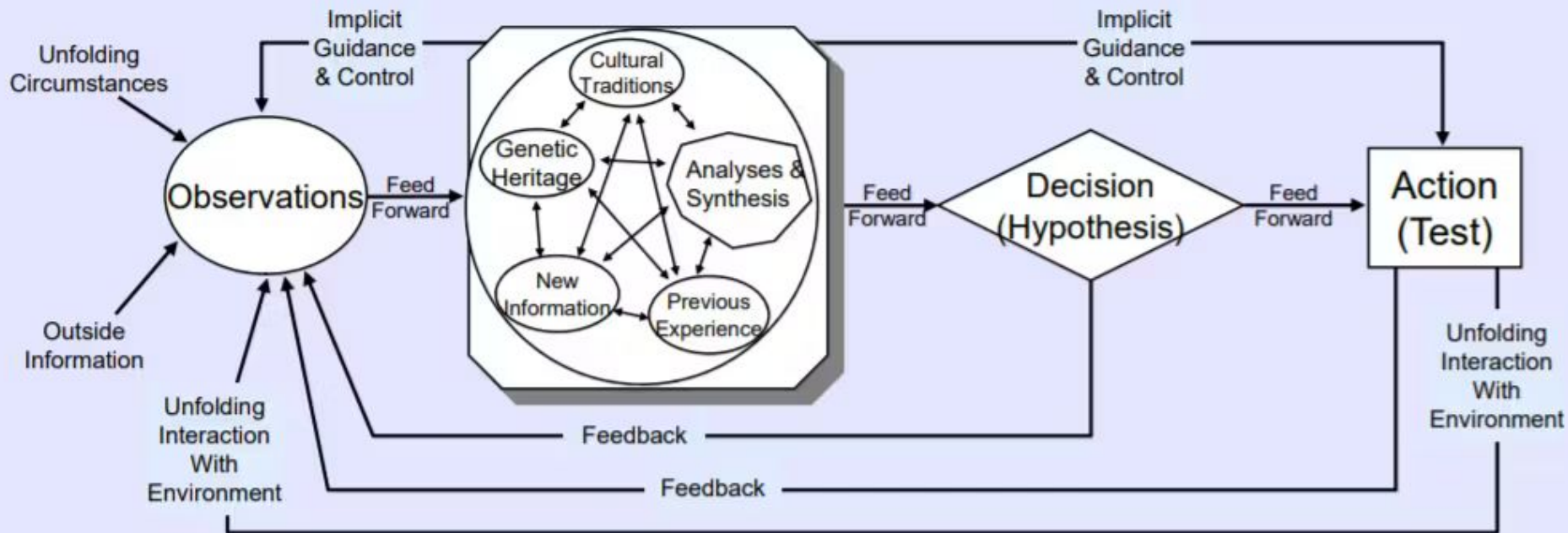
The OODA Loop

Observe

Orient

Decide

Act



The Lore



Where Data Engineering Loops Break

Software Engineer



Data Engineer



Compounding Costs

5
min

to fix in dev

5
hrs

to fix in prod

5
wks

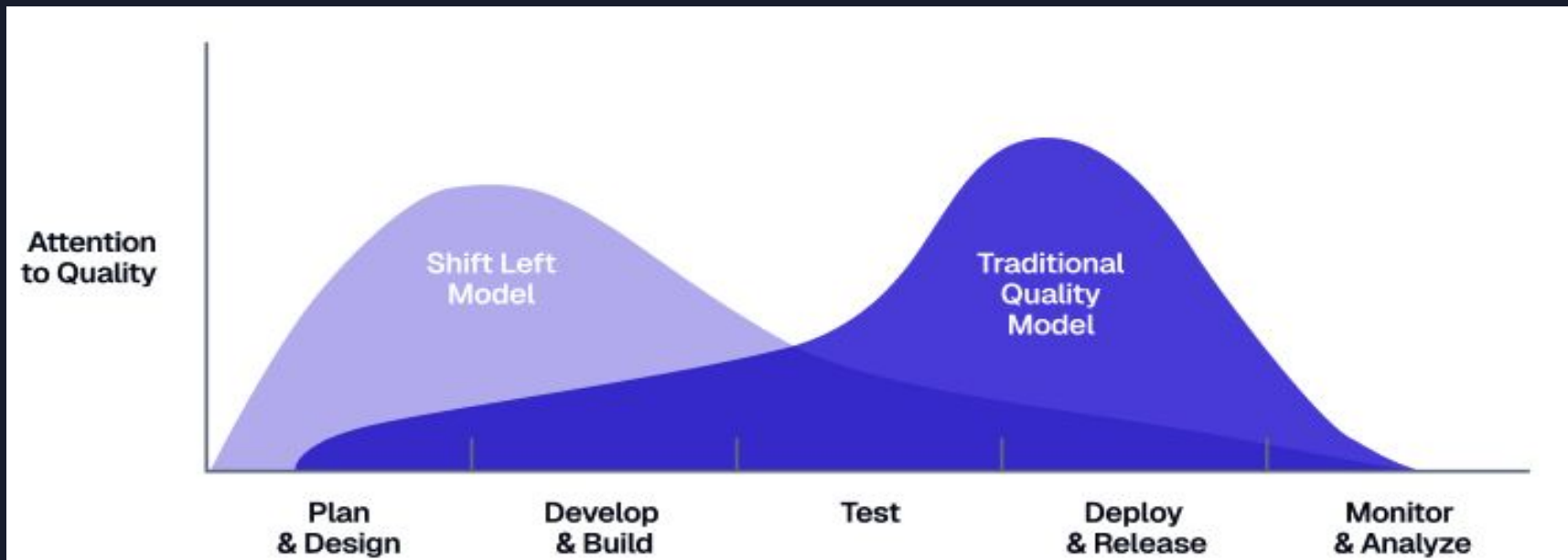
To rebuild trust

Trust is the most important part of data
Everything must support that end

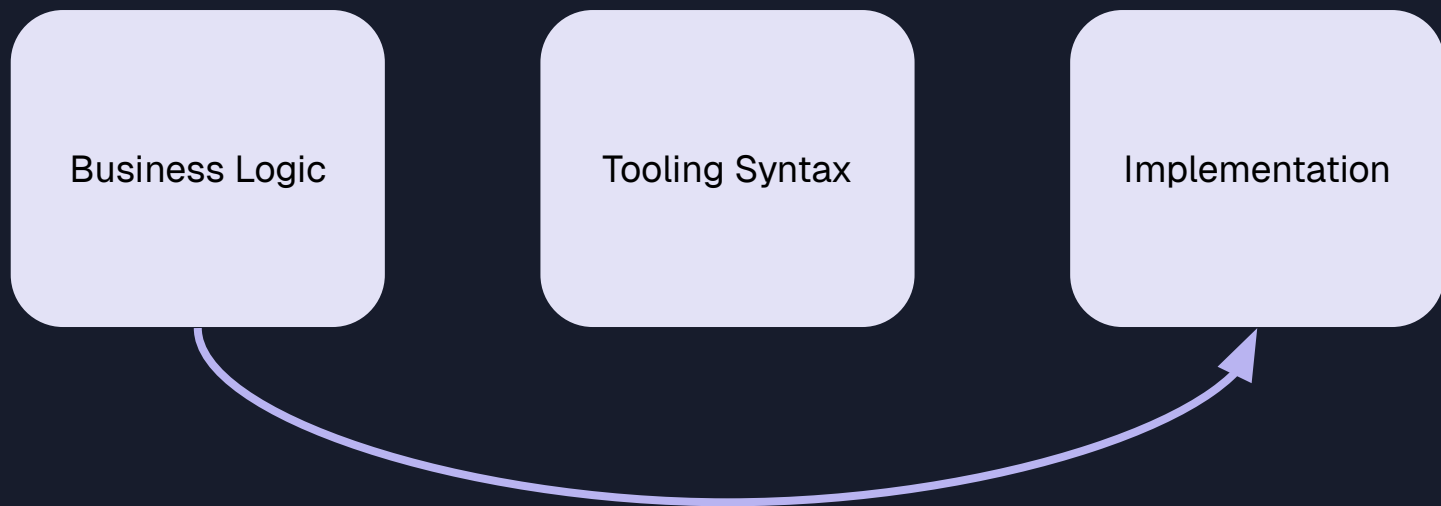
So Called Developer Experience

Day 1 Operations

Shifting Left

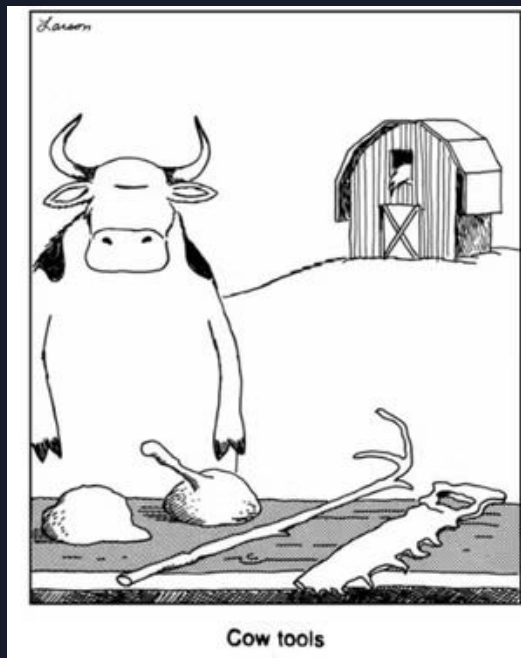


Abstractions that Match The Mental Model



AI-Assisted Development

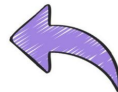
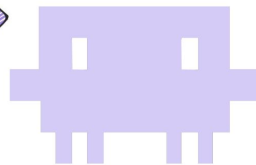
- Some previously challenging problems are now trivial
- Other problems have become massive bottlenecks
- Golden age of personal software



How it feels to use someone else's setup

**Our Claude plugin combines
domain knowledge about Dagster
best practices with the internal
coding standards we use**

Domain Knowledge



Coding Standards



master



1 Branch



0 Tags

Go to file



Add file



<> Code



cmpadden

Merge pull request #7 from dagster-io/colton/feat/remote-dbt-recommen...



a3ecd2c · last week



33 Commits



.claude-plugin

Ensure yes is piped into create- commands; refactor fold...

last week



.github

Add .github/PULL_REQUEST_TEMPLATE.md

last week



plugins

Update convention to recommend remote dbt project con...

last week



README.md

Restructure README.md by plugin

last week



README



Claude Plugins for Dagster

A collection of Claude Code plugins for building workflows and data pipelines using Dagster.

Installation

Add this marketplace to Claude Code:

```
/plugin marketplace add dagster-io/dagster-claude-plugins
```



Install a plugin:

```
/plugin install dg@claude-plugins-dagster
```



To update your marketplace plugins:

```
/plugin marketplace update
```



Platform Management

Day 2 Operations



Presidential Alert
THE GIRLS ARE FIGHTINGGGG

Alert Fatigue

Principles for effective alerting:

- Alert on symptoms that require human judgment, not every anomaly
- Severity levels that actually mean something (if everything is critical, nothing is)
- Alerts with context: what broke, what's affected, where to start investigating



System Orientation & Observability

What good orientation looks like:

- Visual lineage: trace from symptom to root cause
- Run history: what changed recently?
- Asset health at a glance, not buried in 15 dashboards

Data Quality!



Why data quality falls through the cracks:

- Treated as an afterthought ("we'll add checks later")
- Siloed in separate tools that nobody checks
- No one's explicit responsibility

Shifting left on data quality:

- Embed checks in the pipeline, not downstream
- Fail fast when expectations aren't met
- Make quality visible to everyone, not just data engineers

How Come we never talk about this?

Data Team Culture

Some things never change

- It could all be so simple
- Tech debt?
- Prioritization & leadership
- Its easier to complain and argue about technical issues then work through organizational ones

Organizational Challenges

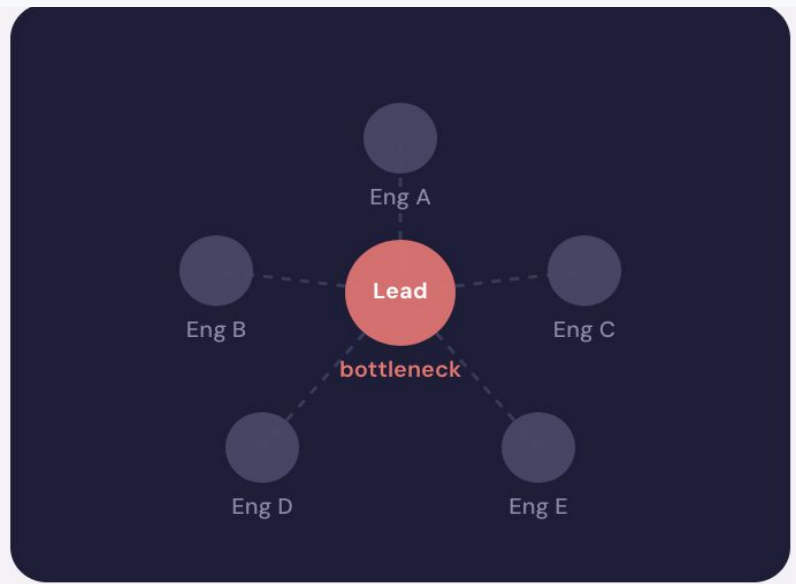
Biggest Bottlenecks

Respondents were asked to identify the single biggest bottleneck in their data organization. The results reveal that organizational and process issues outweigh purely technical challenges.

BOTTLENECK	PERCENTAGE
Legacy systems / technical debt	25.4%
Lack of leadership direction	21.3%
Poor requirements / upstream issues	18.8%
Talent / hiring challenges	11.4%
Data quality issues	10.1%
Compute costs	5.2%
Tool complexity	2.7%

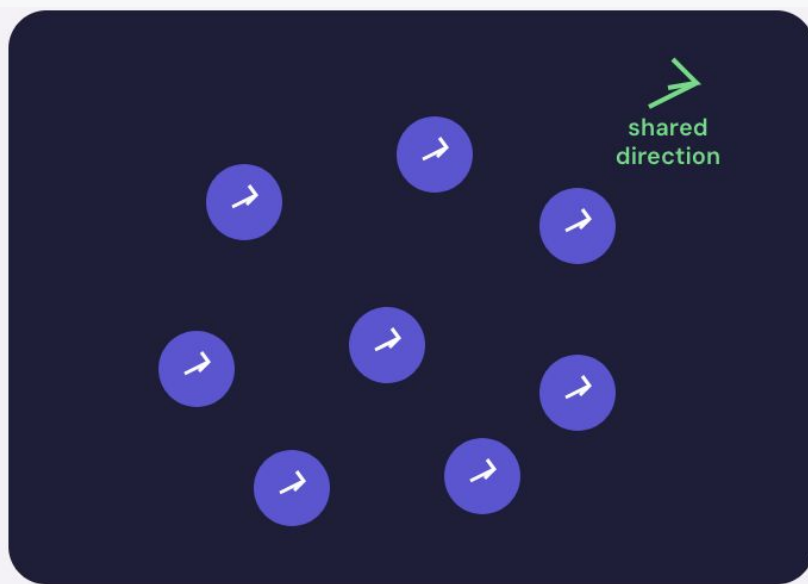
Centralized Bottleneck

Every decision flows through the single point of failure.



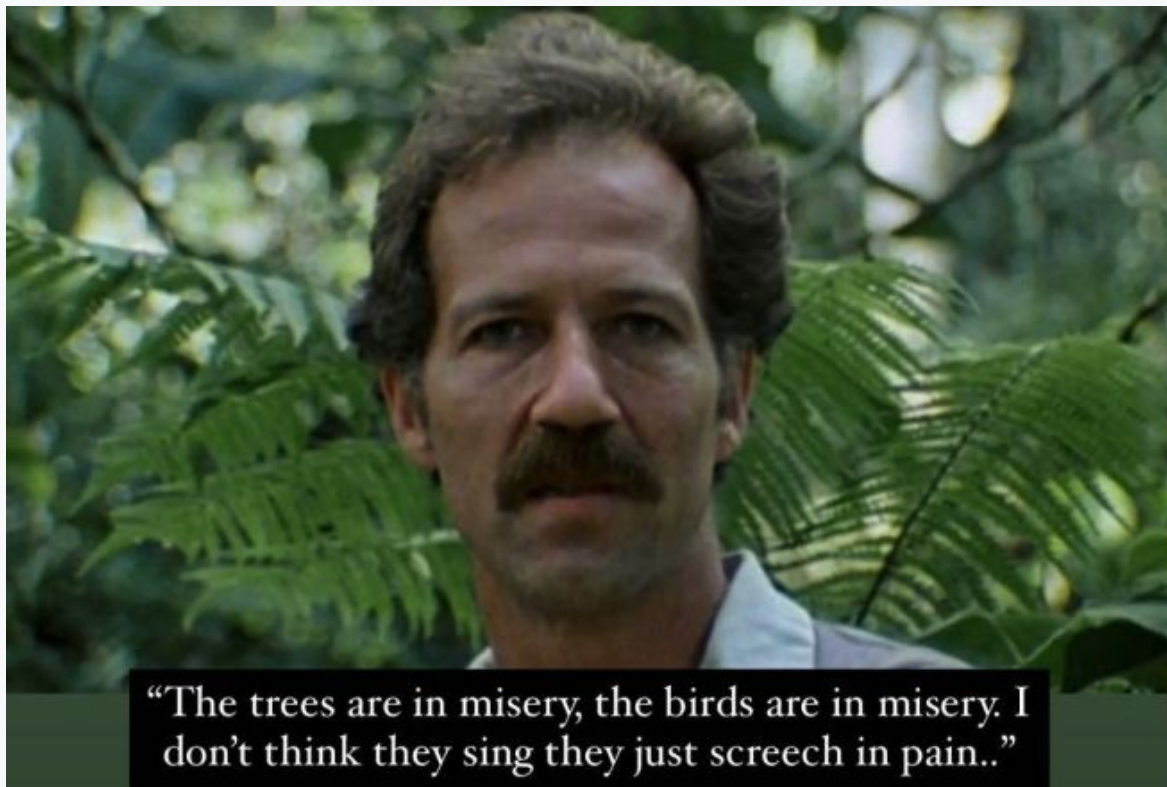
Schwerpunkt

Shared understanding of mission & priorities.
Everyone moves independently in the same direction.



Tribal Knowledge

- Runbooks? Never heard of them
- How fast can new team membership?



“The trees are in misery, the birds are in misery. I don’t think they sing they just screech in pain..”

Your Team Matters

- Team Composition
- Retention!
- Militant in defending your culture

General Interference with Organizations and Production

(1) Insist on doing everything through "channels." Never permit short-cuts to be taken in order to expedite decisions.

(2) Make "speeches." Talk as frequently as possible and at great length. Illustrate your "points" by long anecdotes and accounts of personal experiences. Never hesitate to make a few appropriate "patriotic" comments.

(3) When possible, refer all matters to committees, for "further study and consideration." Attempt to make the committees as large as possible — never less than five.

(4) Bring up irrelevant issues as frequently as possible.

(5) Haggle over precise wordings of communications, minutes, resolutions.

(6) Refer back to matters decided upon at the last meeting and attempt to re-open the question of the advisability of that decision.

(7) Advocate "caution." Be "reasonable" and urge your fellow-conferes to be "reasonable" and avoid haste which might result in embarrassments or difficulties later on.

(8) Be worried about the propriety of any decision — raise the question of whether such action as is contemplated lies within the jurisdiction of the group or whether it might conflict with the policy of some higher echelon.

CORPORATE
REBELS

Platform Management & Culture

Evaluating Your Data Platform

Feedback Loop Audit

Question	Healthy	Unhealthy
How long from code change to knowing if it works?	Minutes	Hours/Days
When something breaks, how long to understand why?	Minutes	Hours of digging
How do you know if data is correct?	Automated checks	Someone complains
Can a new team member ship in their first week?	Yes	"It's complicated"
Do alerts lead to action?	Usually	Mostly noise
Does your team have a shared understanding of priorities?	Clearly articulated	Implied/assumed
How much critical knowledge lives only in someone's head?	Minimal, documented	A lot, and everyone knows it

Prioritization

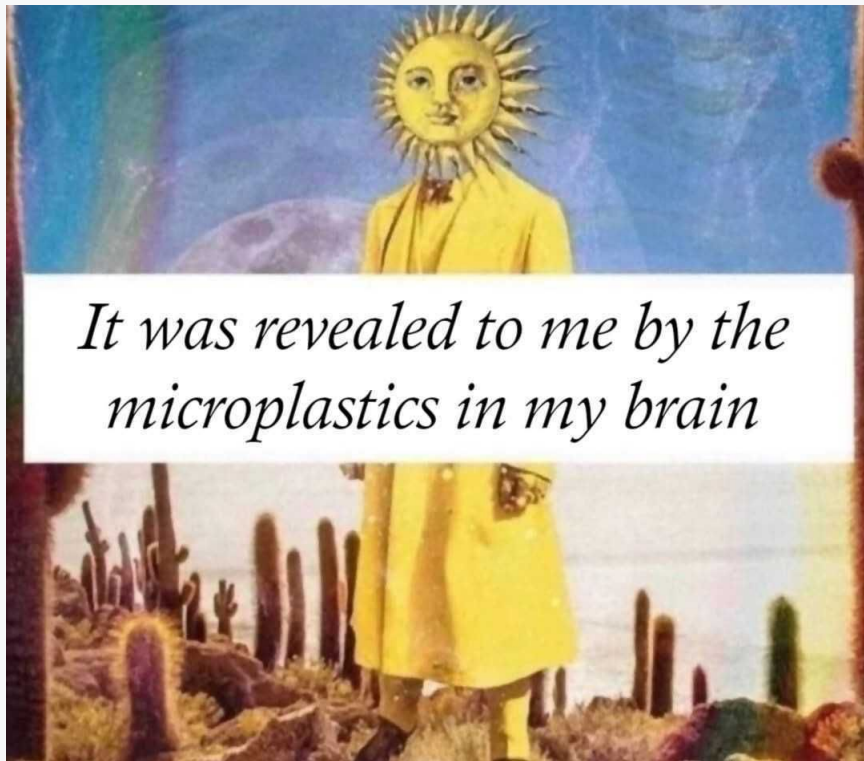
Which feedback loop is the most broken?

Observe broken? You're not seeing problems at all. Fix alerting and monitoring.

Orient broken? You see problems but can't understand them. Fix lineage, documentation, and shared context.

Decide broken? You understand problems but can't agree on what to do. Fix culture, priorities, and Schwerpunkt.

Act broken? You know what to do but can't do it fast enough. Fix tooling, deployment, and abstractions.



Demo



Key Points

Feedback loops are the multiplier

Fast, high-quality feedback turns average engineers into highly productive ones. Slow, noisy feedback does the opposite. You're not fighting your tools, you're fighting entropy.

Day 1 and Day 2 are connected, culture is the bridge

The abstractions you choose during development determine how manageable operations will be. The culture you build determines whether your team actually uses the feedback loops you've created.

Next Steps

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Thank You!

